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Total No. of Questions : 05]

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EYS-88

B. Tech. (CSE. IT & Elect.)

Examination, 2024

(Second Semester)

ENGINEERING GRAPHICS & CAD

BE-205

Time : 3 Hours]

[Maximum Marks : 60

Note : (i) Attempt all the questions. Part (a) of each question is compulsory and Part (b) has an internal choice.

(ii) All questions carry equal marks. Part (a) of each question carries 4 marks and Part (b) carries 8 marks.

(iii) Construct neat and clean, well labeled diagram.

1. (a) Differentiate between the following : 4
- (i) Enlarging Scale and Decreasing Scale
 - (ii) Epicycloid and Hypocycloid

- (b) Construct a scale of 1 : 4 to show centimeters and long enough to measure up to 5 decimeters. 8

Or

The major and minor axis of an ellipse are 140 mm and 90 mm respectively. Find the foci, draw the ellipse using arc of circle method. Also draw the tangent and normal on it.

- (a) What do you mean by orthographic projection ? 4
- (b) Draw the projection of the following points on the same ground line keeping the projectors 25 mm apart : 8
- (i) 'A' in the H. P. and 20 mm behind the V.P.
 - (ii) 'B' 40 mm above the H. P. and 25 mm in front of the V.P.
 - (iii) 'C' 30 mm below the H. P. and 40 mm in front the V.P.
 - (iv) 'D' 25 mm below the H. P. and 25 mm behind the V.P.

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Or

A hexagonal plane of side 50 mm side has an edge in the V.P. The surface of the plane inclined at 35° to the V.P. and the edge on which it rests is inclined at 45° to the H.P. Draw its projections.

3. (a) Explain the following : 4
- (i) Tetrahedron
 - (ii) Regular Pentagon
 - (iii) Right Circular Cone
 - (iv) True Shape
- (b) A pentagonal prism of side of base 25 mm and 60 mm long axis is freely suspended from a corner of the top base. Draw the projections of the prism in suspended position. 8

Or

A right circular cylinder, diameter of base 50 mm and height 80 mm, rests on its base rim such that its axis is inclined at 45° to the H.P. and is parallel to the V.P. A section plane parallel to the H.P. cuts the axis at a distance of 50 mm from its base. Draw its front view and sectional top view.

4. (a) Explain principle of isometric projection. 4
- (b) Draw the isometric view of a square prism of base side 40 mm and axis 60 mm resting on the HP on the : 8
- (i) Base with axis perpendicular to the HP
 - (ii) Rectangular face with axis perpendicular to the VP

Or

Develop the lateral surface of a right regular hexagonal prism of 20 mm base edge and 50 mm height.

5. (a) Write brief notes on the following : 4
- (i) CAD
 - (ii) CAM
- (b) Write about prototype development as a part of design process. 8

Or

What are the various input and output devices more commonly employed for general graphics applications ? Explain in brief.